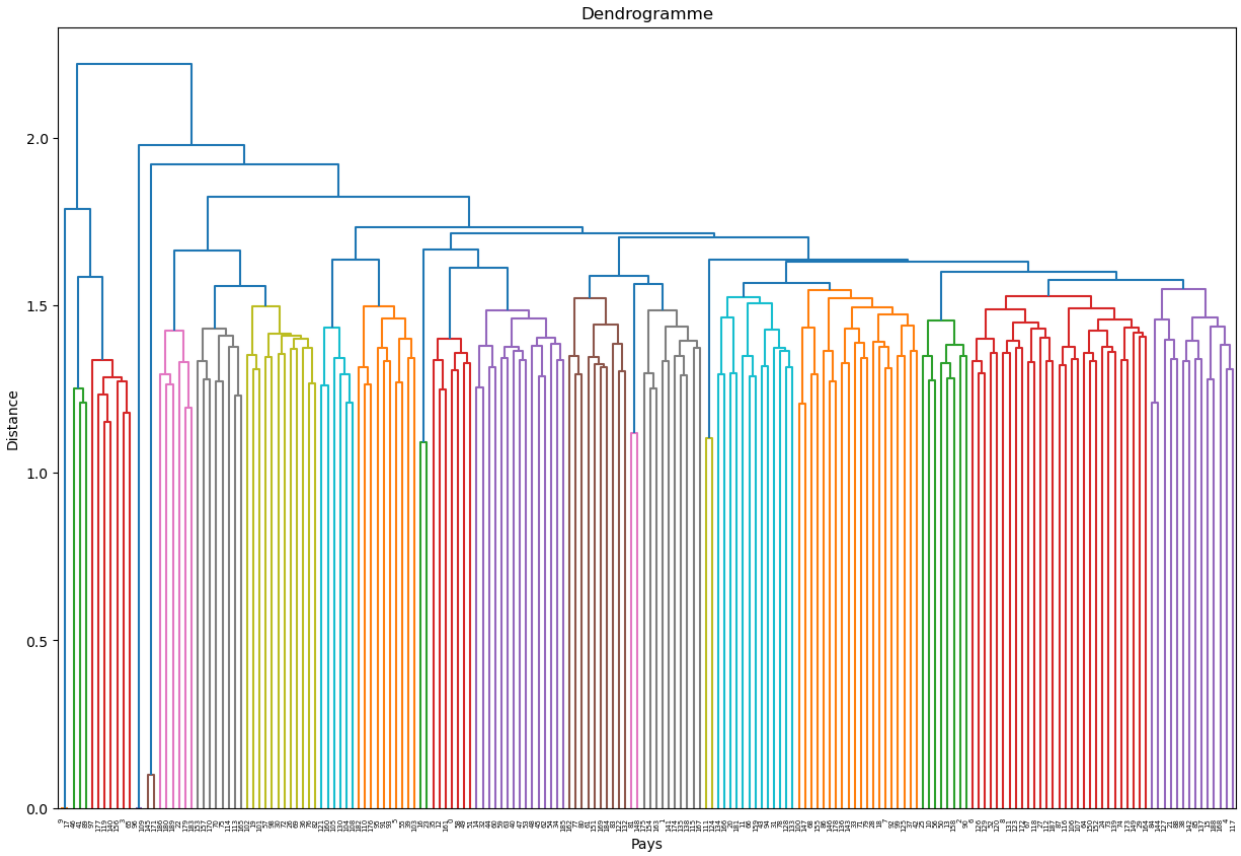
a. Appliquer l'algorithme de partitionnement hiérarchique

```
import warnings

warnings.filterwarnings('ignore')

import pandas as pd
import numpy as np
import re
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage
import matplotlib.pyplot as plt

# Charger les données
df = pd.read_csv('./anthems.csv')

# Nettoyage superficiel
def clean_text(text):
    text = text.lower()
    text = re.sub(r'<.*?>', '', text)
    text = re.sub(r'[^a-zA-Z\s]', '', text)
    text = text.strip()
    return text

df['Anthem'] = df['Anthem'].apply(clean_text)

# Vectorisation
vectorizer = TfidfVectorizer(stop_words='english')
X = vectorizer.fit_transform(df['Anthem'])

# Convertir la matrice sparse en une matrice dense
X_dense = X.toarray()

# Appliquer l'algorithme de partitionnement hiérarchique
hierarchical_clustering =
AgglomerativeClustering(distance_threshold=0, n_clusters=None)
hierarchical_clustering.fit(X_dense)

# Tracer le dendrogramme
plt.figure(figsize=(15, 10))
dendrogram(linkage(X_dense, method='ward'))
plt.title('Dendrogramme')
plt.xlabel('Pays')
plt.ylabel('Distance')
plt.show()
```

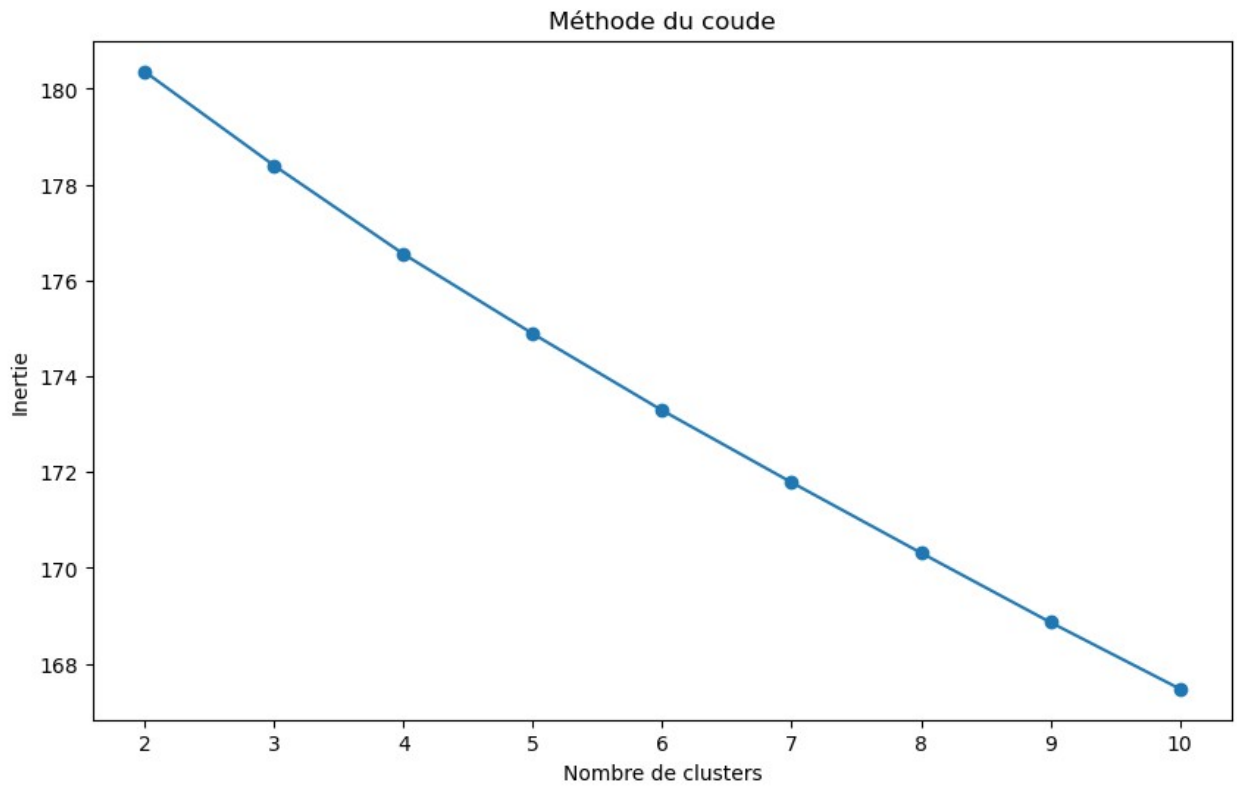


b. Déterminer le nombre optimal de clusters par la méthode du coude

Nous allons tracer le dendrogramme pour visualiser les distances et déterminer le nombre optimal de clusters.

```
import matplotlib.pyplot as plt
from scipy.cluster.hierarchy import linkage, fcluster
from scipy.spatial.distance import pdist

# Calculer la matrice de linkage
Z = linkage(X_dense, method='ward')

# Calculer les inerties pour différentes valeurs de k
inertias = []
K = range(2, 11)

for k in K:
    # Créer les clusters
    labels = fcluster(Z, k, criterion='maxclust')

    # Calculer l'inertie
    inertia = 0
    for cluster in range(1, k+1):
        cluster_points = X_dense[labels == cluster]
```

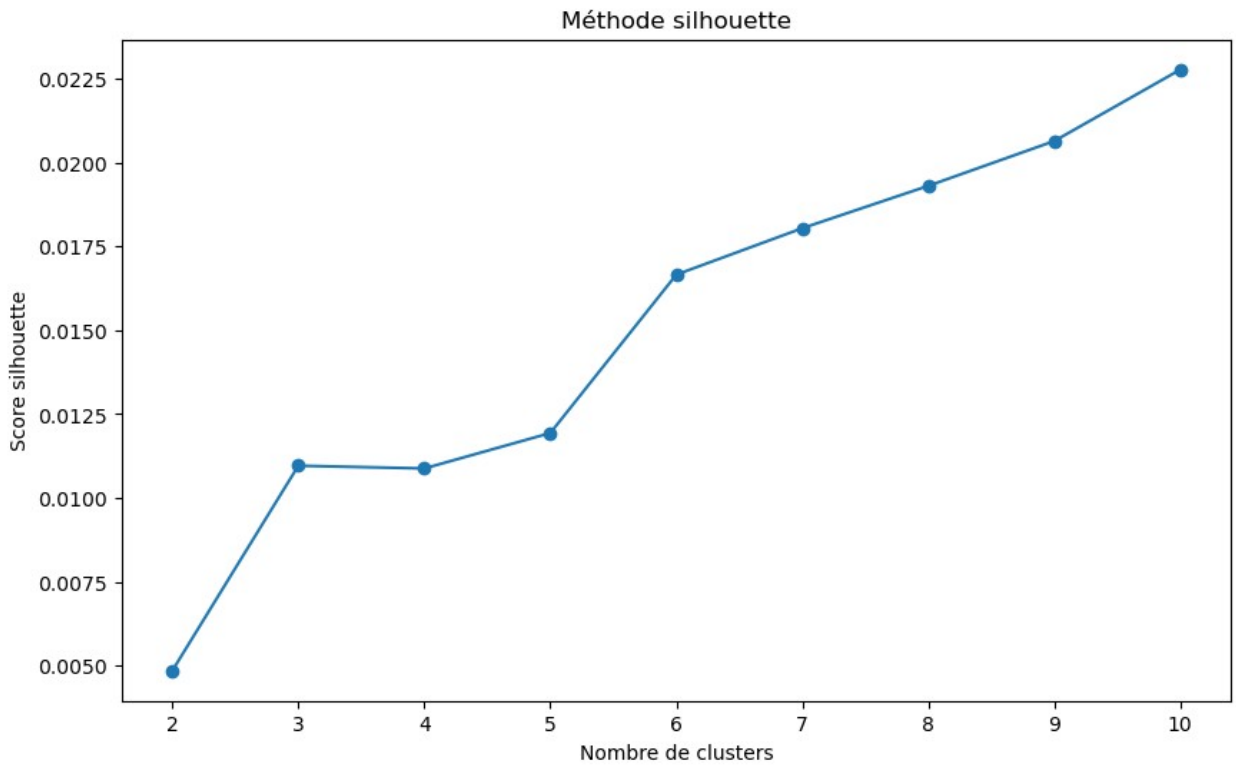
```
        centroid = cluster_points.mean(axis=0)
        inertia += ((cluster_points - centroid) ** 2).sum()

    inertias.append(inertia)

# Tracer la courbe de la méthode du coude
plt.figure(figsize=(10, 6))
plt.plot(K, inertias, marker='o')
plt.title('Méthode du coude')
plt.xlabel('Nombre de clusters')
plt.ylabel('Inertie')
plt.show()
```



c. Déterminer le nombre optimal de clusters par la méthode silhouette

```
from sklearn.metrics import silhouette_score

# Méthode silhouette
silhouette_scores = []
for k in range(2, 11):
    hierarchical_clustering = AgglomerativeClustering(n_clusters=k)
    labels = hierarchical_clustering.fit_predict(X_dense)
    score = silhouette_score(X_dense, labels)
    silhouette_scores.append(score)
```

```
plt.figure(figsize=(10, 6))
plt.plot(range(2, 11), silhouette_scores, marker='o')
plt.title('Méthode silhouette')
plt.xlabel('Nombre de clusters')
plt.ylabel('Score silhouette')
plt.show()
```



d. Ajouter au dataset original une variable “cluster”

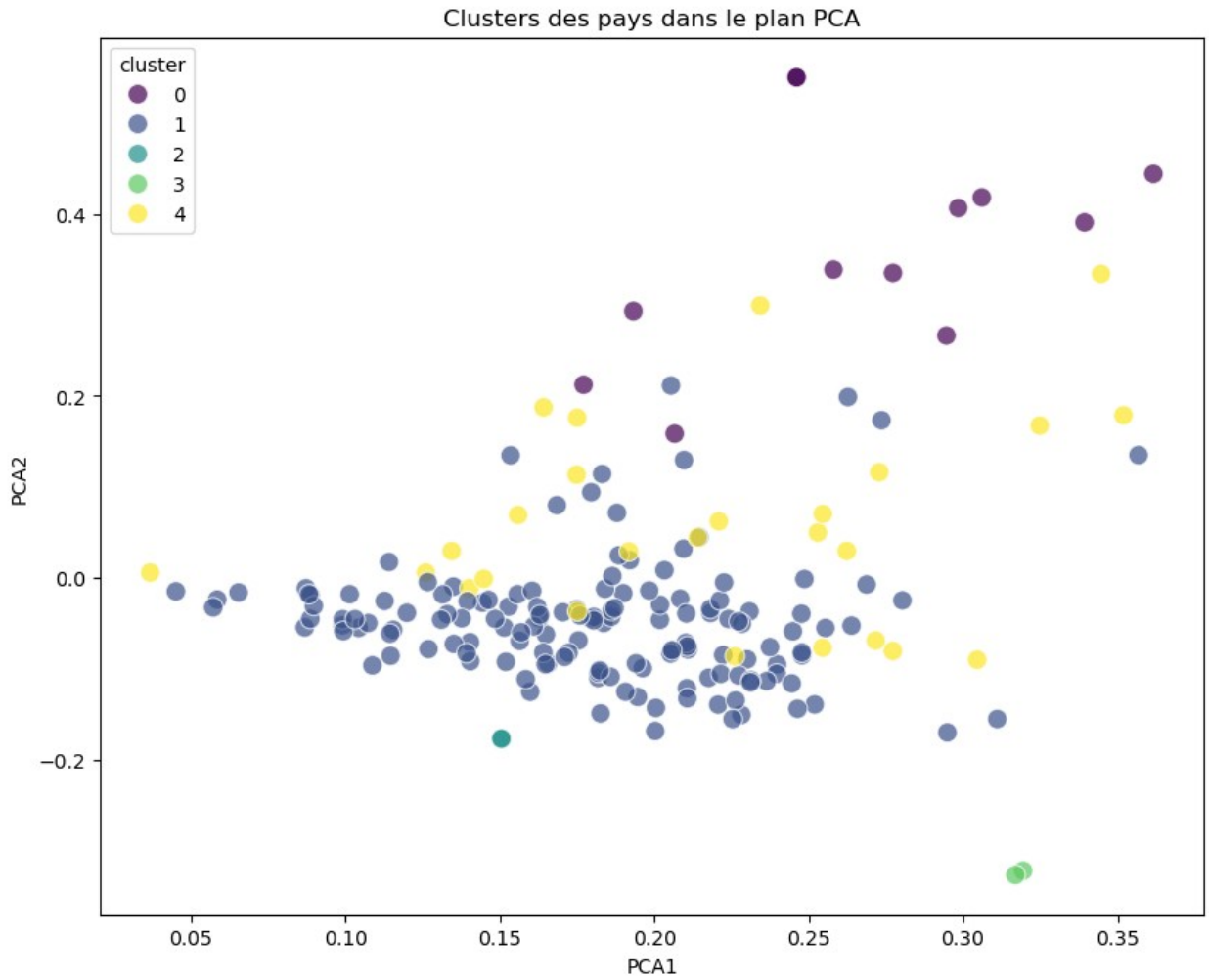
```
# Supposons que le nombre optimal de clusters est k=5
optimal_k = 5
hierarchical_clustering =
AgglomerativeClustering(n_clusters=optimal_k)
clusters = hierarchical_clustering.fit_predict(X_dense)

# Ajouter les clusters au dataset original
df['cluster'] = clusters

# Sauvegarder le dataframe avec les clusters
df.to_csv('./anthems_with_clusters_hierarchical.csv', index=False)

# Vérifier les premières lignes du dataframe avec les clusters
print(df.head())
```

```
      Country Alpha-2 Alpha-3 Continent  \
0     Albania      AL     ALB    Europe
```

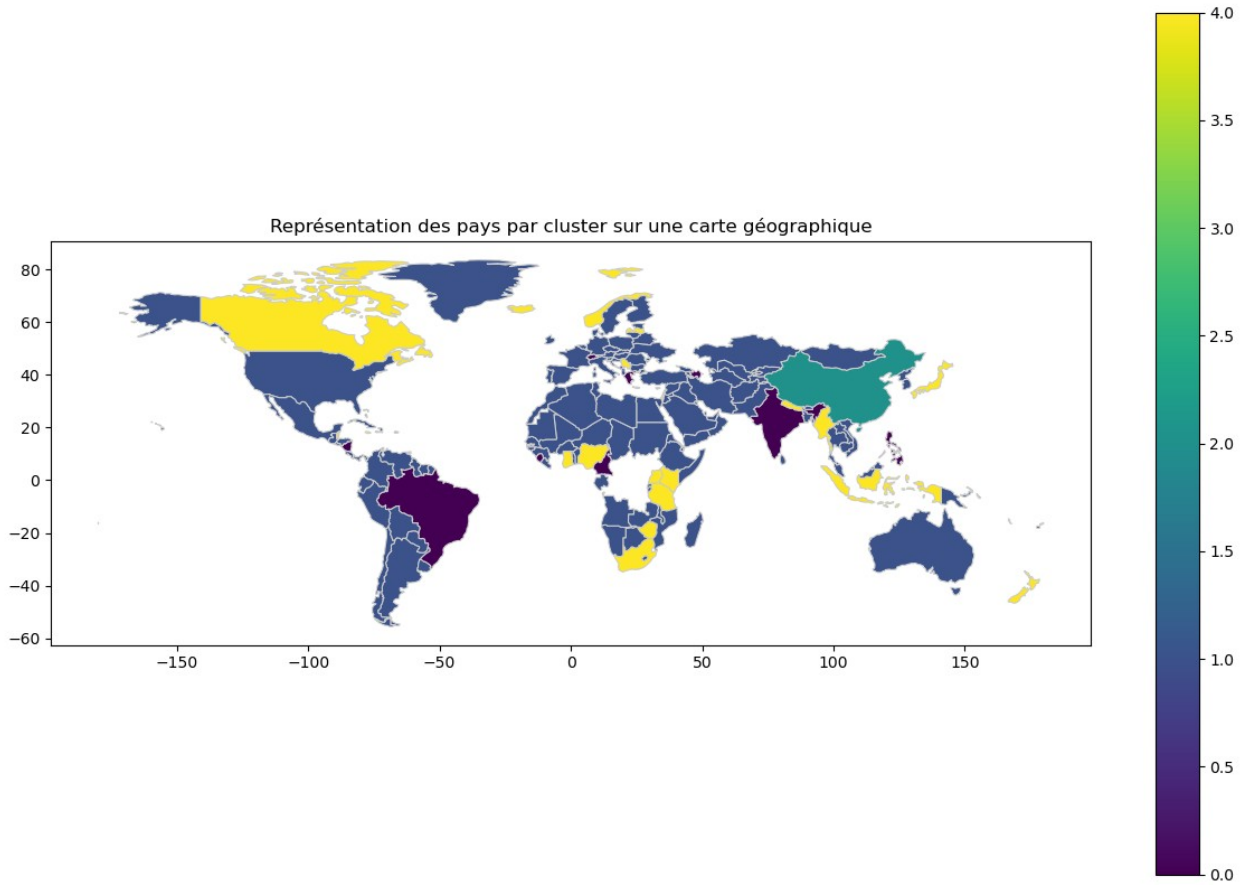
```
1     Armenia      AM     ARM    Europe
2     Austria      AT     AUT    Europe
3  Azerbaijan      AZ     AZE    Europe
4     Belarus      BY     BLR    Europe

                                                Anthem  cluster
0  around our flag we stand united with one wish ...        1
1  our fatherland free independent that has for c...        1
2  land of mountains land by the river land of fi...        1
3  azerbaijan azerbaijan the glorious fatherland ...        0
4  we belarusians are peaceful people wholehearte...        1
```

e. Représenter les pays dans le plan engendré par les deux premières composantes principales

```
from sklearn.decomposition import TruncatedSVD
import seaborn as sns
import matplotlib.pyplot as plt

# Réduction de dimension avec TruncatedSVD
svd = TruncatedSVD(n_components=2)
components = svd.fit_transform(X_dense)

# Ajouter les composantes principales au dataframe
df['PCA1'] = components[:, 0]
df['PCA2'] = components[:, 1]

# Visualisation des clusters dans le plan PCA
plt.figure(figsize=(10, 8))
sns.scatterplot(x='PCA1', y='PCA2', hue='cluster', data=df,
palette='viridis', s=100, alpha=0.7)
plt.title('Clusters des pays dans le plan PCA')
plt.show()
```




```
import geopandas as gpd
import matplotlib.pyplot as plt

# Charger une carte du monde
world = gpd.read_file(gpd.datasets.get_path('naturalearth_lowres'))

# Fusionner avec le dataframe contenant les clusters
merged = world.set_index('name').join(df.set_index('Country'))

# Représenter les pays avec une couleur par cluster
fig, ax = plt.subplots(1, 1, figsize=(15, 10))
merged.plot(column='cluster', cmap='viridis', linewidth=0.8, ax=ax,
edgecolor='0.8', legend=True)
plt.title('Représentation des pays par cluster sur une carte
géographique')
plt.show()
```



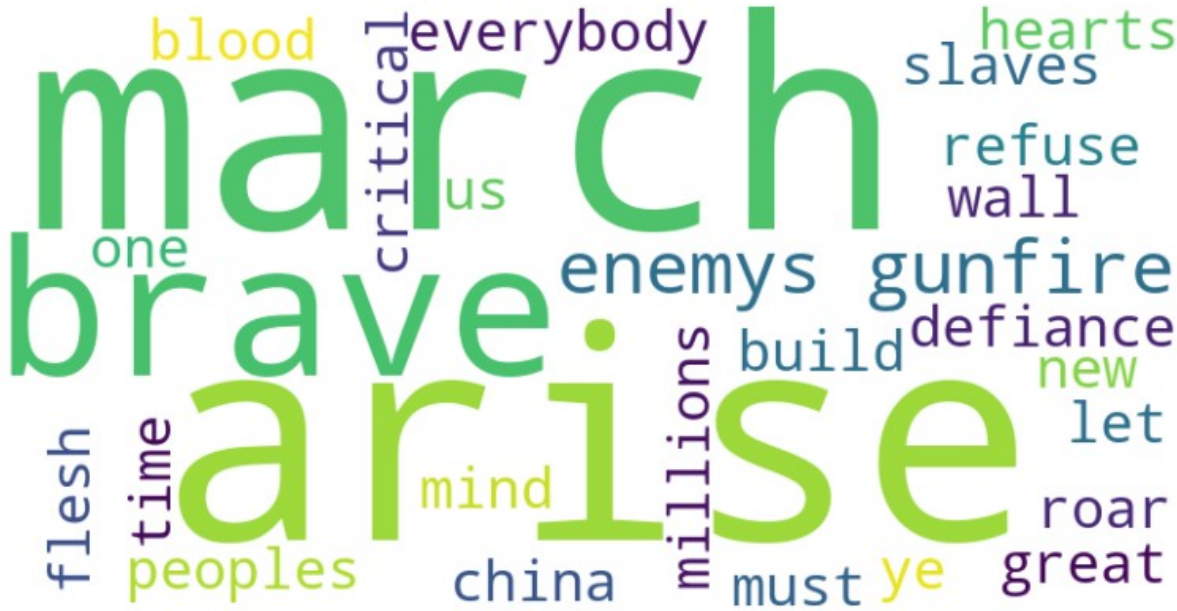
g. Représenter les mots les plus fréquents dans chaque cluster avec un nuage de mots

```
from wordcloud import WordCloud

# Créer des nuages de mots pour chaque cluster
for cluster in range(optimal_k):
    cluster_texts = df[df['cluster'] == cluster]
['Anthem'].str.cat(sep=' ')
    wordcloud = WordCloud(width=800, height=400,
background_color='white').generate(cluster_texts)

    plt.figure(figsize=(10, 5))
    plt.imshow(wordcloud, interpolation='bilinear')
    plt.axis('off')
    plt.title(f'Nuage de mots pour le cluster {cluster}')
    plt.show()
```

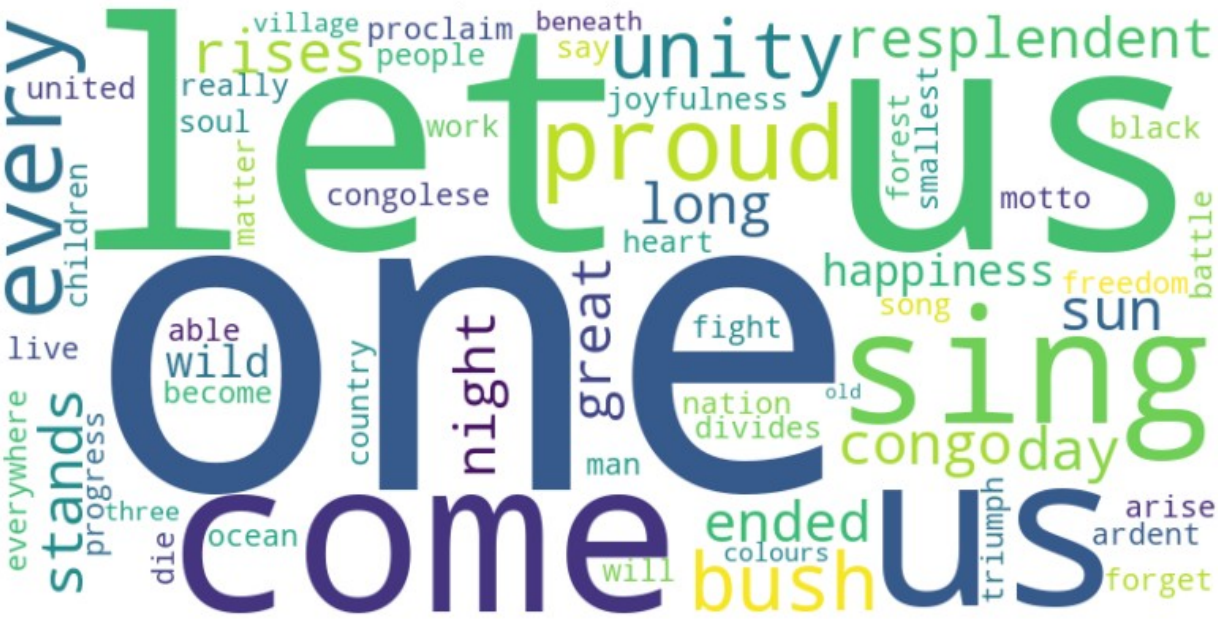
Nuage de mots pour le cluster 0

thy
thou
land
thine
thee
love
hail
light
brazil
glorious
son
may
stand
fatherland
peace
hearts
liberty
art
azerbaijan
beloved
glory
hills
sky
us
god
beautiful
sierra leone
homeland
will
mother
sun
father
name
knew
hast
eyes
bosom
ready
become
people
dwelleth
shed
thousand
uphold
tricolor
now
dispenser
restored
banner
understand
clouds
pledge
mighty
honor
graves
homeland
valor
prevail
live
past
new
hope
feel
idolized
ground
fields
never
might
greet
amongst
shores
blood
others
oh
cause
die
radiance
earned
voice
old
battle
future
pray
slain
lord
high
greatness
blessing
morning
cradle
blissfully
life
sea
skies
one
deep
praise
holy
faith
descend
sacrifice
bright
intense
valorous
victory
joy
art
true
adored
shining
beloved
divinely
devotion
valley
dear
sweeping
sword
honour
strong
destiny
death
beyond
kind
proud
coast
storied
shone
plains
folk
appereth
lipiranga
enshroud
cleansing
sunset
near
departing
arm
rosegarden
keep
loving
dream
smiling
chest
power
country
gloom
indias
limpid
bravely
place
bliss
yet
died
mist
defended
dwellest
placid
managed
thoult
annals
dawn
want
sound
made
heroes
warrior
head
foe
well
bless
gave
raise
past
noble
sublime
shine
golden
living
without
justice
call
witness
way
fight
right
men
king
time
hail
soldier
cause
sacred
battle
heaven
thy
women
roar
praise
lead
war
earth
spirit
prosperity
will
fulfill
progress
swear
people
brother
honour
blessed
take
independence
free
arise
upon
yoke
father
see
arm
children
pride
eternal
work
ancestor
banner
soul
star
shore
spain
precious
motherland
symbol
remain
protect
path
rise
mountain
wave
island
good
stand
come
last
valley
chain
old
field
power
give
sing
oh
song
make
day
every
great
towards
tear
death
joy
march
proudly
sky
voice
blue
cry
soil
know
let
us
long
beautiful
country
heart
nation
unity
homeland
happiness
hand
light
man
flag
love
now
strength
name
always
world
thee
proud
sun
high
hill
never
future
dear
faith
god
home
brave
forever
slave
glorious
strong
heroes
warrior
head
mother
place
tyrant
break
must
live
victory
long
struggle
dignity
symbol
fate
freedom
glory
fatherland
liberty
africa
hope
oer
shout
eye
union
immortal
defend
truth
word
beauty
ancient
river
happy
bright
today
green
new
keep
sword
thou
power
tear
towards
proudly
blessed
take
independence
spain
yoke
upon
precious
roar

Nuage de mots pour le cluster 2



Nuage de mots pour le cluster 3



Nuage de mots pour le cluster 4

land
thee
us
thy
god
may
love
nation
free
stand
bless
peace
canada
africa
freedom
guard
mountain
indonesia
country
let
heart
call
hope
great
native land
unity
tanzania
serbian
children
liberty
lord
defend
one
forever
make
people
sea
true
justice
thousand
years
motherland
o
oh
truth
together
pledge
build
faith
duty
new zealand
father
right
south
defend
new
tears
ghana
race
teach
care
victory
mother
uphold
serbian
icelands
thousand
latvia
high
labour
rugged
union
keep
strength
whose
serve
hear
life
blue
might
command
voices
firm
north
true north
perish
rise
prosper
past
united
sun
glory
praise
ocean
will
patriot
prairies
reign
day
heritage
prayer
homage
give
fame
oer
fill
dear
rising
find
vow
fatherland
arise
sovereign
hold
great indonesia
strong
advance
north strong
fair
destiny
leader
years
cherish
bring
courage
heaven
guide
slavery
wisdom
still
equal
battle
daughters
cause
fertile
steadfast
stormscarred
jamaica jamaica
western
forever
hour
old
night
face
rivers
unto
need
hard
star
lye
yet
never
within
home
strife
pray
spread
brave
sublime
latvian
power
crown
blood
grace
bond
wealth
live
hail
sons
sires
safeguard
well
increase
grow
age
beloved
sing
help
send
strengthen
flag
joy
east
thou
thank
almighty protect
place
every creed
host
defender
grant
spirit
oh uganda
hundred
serbian race
mighty
build
name
fields
sovereign
hold
truth
pledge
together
faith
duty